Calculus II

Tutorial #12 12–16 May 2025

Topics:

- The method of undetermined coefficients (continued)
- The method of variation of-parameters
- Series solutions to differential equations (a self-study)

• **True-False Review**. Decide if the given statement is **True** or **False**, and justify your answer.

1. If $p(x)$ and $q(x)$ are polynomials, then the differential equation
$$y'' + p(x)y' + q(x)y = 0$$
has no singular points.

2. (a self-study) The radius of convergence of a power series solution to the differential equation
$$y'' + \frac{1}{x}y' + \frac{1}{x+1}y = 0$$
centered at $x = -3$ is at most 2.

3. (a self-study) If $\displaystyle\sum_{n=0}^{\infty} a_n x^n$ is a poser series solution to the differential equation
$$y'' + p(x)y' + q(x)y = 0,$$
then so is $\displaystyle\sum_{n=0}^{\infty} a_n x^{n+1}$.

• **Problems**.

4. Using the method of undetermined coefficients, find the general solution to the given differential equation.
   (a) $y'' + 4y' + 4y = 5xe^{-2x}$.
   (b) $y'' - 5y' = -5x^2 + 2x$.

In Problems 5–7, using the method of variation-of-parameters, find the general solution to the given differential equation.

5. (a) $y'' + 6y' + 9y = \dfrac{2e^{-3x}}{x^2+1}$

   (b) $y'' - 4y = \dfrac{8}{e^{2x}+1}$

6. (a) $y'' - 4y' + 5y = e^{2x}\tan x, \quad 0 < x < \pi/2$

   (b) $y'' - 6y' + 9y = 4e^{3x}\log x, \quad x > 0$

7. (a) $y'' + y = \sec x + 4e^x, \quad |x| < \dfrac{\pi}{2}.$

   (b) $y'' + y = \csc x + 2x^2 + 5x + 1, \quad 0 < x < \pi$

8. (A self-study) Find the series solutions to the given linear differential equation.

   (a) $y' - y = 0.$

   (b) $(1-x^2)y'' - 2xy' + 2y = 0.$

   (c) $y'' + y' + x^2 y = 0.$

For Problems 9–10, determine terms up to and including x^5 in two linearly independent power series solution of the given differential equation. State the radius of convergence of the series solutions.

9. (A self-study) $y'' + xy' + (2+x)y = 0.$

10. (A self-study) $y'' - e^x y = 0.$

END

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Answers to selected problems

1. True

2. False

3. False

4. (a) $y(x) = C_1 e^{-2x} + C_2 x e^{-2x} + \dfrac{5}{6}x^3 e^{-2x}$, (b) $y(x) = C_1 + C_2 e^{5x} + \frac{1}{3}x^3$

5. (a) $y(x) = e^{-3x}[C_1 + C_2 x + 2x\arctan x - \ln(x^2+1)$ (b) $y(x) = C_1 e^{2x} + C_2 e^{-2x} + [\ln(e^{2x}+1) - 2x - e^{-2x}]e^{2x} - \ln(e^{2x}+1)e^{-2x}$

6. (a) $y(x) = (C_1 \cos x + C_2 \sin x)e^{2x} + e^{2x}[\cos x - \ln|\sec x + \tan x|]$ (b) $y(x) = C_1 e^{3x} + C_2 x e^{3x} + x^2 e^{3x}(2\ln x - 3)$,

7. (a) $y(x) = C_1 \cos x + C_2 \sin x + 2e^x + \cos x \ln(\cos x) + x \sin x$ (b) $y(x) = C_1 \cos x + C_2 \sin x - x\cos x + \ln(\sin x)\sin x + 2x^2 + 5x - 3$

8. $y(x) = a_0 + a_0 x + \dfrac{a_0}{2!}x^2 + \dfrac{a_0}{3!}x^3 + \cdots = a_0 \sum_{n=0}^{\infty} \dfrac{x^n}{n!} = a_0 e^x$ (b) $y(x) = a_1 x + a_0(1 - x^2 - \dfrac{1}{3}x^4 - \dfrac{1}{5}x^6 - \cdots)$ (c) $y(x) = a_0(1 - \dfrac{1}{12}x^4 + \dfrac{1}{60}x^5 + \cdots) + a_1(x - \dfrac{1}{2}x^2 + \dfrac{1}{6}x^3 - \dfrac{1}{24}x^4 - \dfrac{1}{24}x^5 + \cdots)$

9. $y_1(x) = 1 - x^2 - \frac{1}{6}x^3 + \frac{1}{3}x^4 + \frac{11}{120}x^5 + \cdots$, $y_2(x) = x - \frac{1}{2}x^3 - \frac{1}{12}x^4 + \frac{1}{8}x^5 + \cdots$, $R = \infty$

10. $y_1(x) = 1 + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{12} + \frac{x^5}{24} + \cdots$, $y_2(x) = x + \frac{x^3}{6} + \frac{x^4}{12} + \frac{x^5}{30} + \cdots$, $R = \infty$